Converting between Units

Exam code: 0607

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Converting units

How do I convert units of lengths?

- For **metric units**, conversion can be done by simply **multiplying or dividing** by **powers of 10**
- You need to know the **basic conversions** for **length**
  - 1 cm = 10 mm
  - 1 m = 100 cm
  - 1 km = 1000 m
- You need to decide whether to **multiply or divide** by the conversion factor
  - Ask, **does the number of units increase or decrease?**
    - The number of mm is 10 times **bigger than** the number of cm so **mm = cm × 10**
- You can do the conversion in **stages**
  - First convert kilometres into metres then metres into centimetres

How do I convert units of mass?

- For **metric units**, conversion can be done by simply **multiplying or dividing** by **powers of 10**
- You need to know the **basic conversions** for **mass**
  - 1 g = 1000 mg
  - 1 kg = 1000 g
  - 1 tonne = 1000 kg

How do I convert units of volume/capacity?

- For **metric units**, conversion can be done by simply **multiplying or dividing** by **powers of 10**
- You need to know the **basic conversions** for **capacity**
  - 1 litre = 100 cl = 1000 ml
    - 1 cl = 10 ml
- You must know the **basic conversion** between **volume** and **capacity**
  - 1 ml = 1 cm^3
  - 1 litre = 1000 ml = 1000 cm^3
  - 1 m^3 = 1000 litres

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Worked Example

Convert

(a) 54 cm to mm

Answer:

1 cm = 10 mm

54 cm = (54 × 10) mm

540 mm

(b) 12 300 cm to km

Answer:

First convert from cm to m
100 cm = 1 m

12 300 cm = (12 300 ÷ 100) m = 123 m

Now convert from m to km
1000 m = 1 km

123 m = (123 ÷ 1000) km

0.123 km

(c) 485 g to kg

Answer:

1000 g = 1 kg

485 g = (485 ÷ 1000) kg

0.485 kg

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